Lab Activity - Data Warehouse Star Schema for Employee

Dimension Tables

DimEmployee

SQL

```sql
dim_employee
------------
employee_key (PK)
emp_no
first_name
last_name
gender
birth_date
hire_date
age
generation
employment_tenure
```

DimDepartment

SQL

```sql
dim_department
--------------
department_key (PK)
dept_no
dept_name
```

DimTitle

```sql
dim_title
---------
title_key (PK)
title_name
career_level
```

DimManager

```sql
dim_manager
-----------
manager_key (PK)
emp_no
manager_name
```

DimDate

```sql
dim_date
--------
date_key (PK)
full_date
day
month
month_name
quarter
year
fiscal_year
week_no
```

Fact Tables

1. FactEmployeeSalary

Business Purpose

Analyze salary trends, compensation changes, and payroll costs.

Grain

One employee salary record per effective date period.

```sql
fact_employee_salary
--------------------
salary_fact_key (PK)

employee_key (FK)
department_key (FK)
title_key (FK)

effective_from_date_key (FK)
effective_to_date_key (FK)

salary_amount
salary_change_amount
salary_change_percent
employment_days
```

Measures

Measure	Description
salary_amount	Employee salary
salary_change_amount	Increase/decrease from previous salary
salary_change_percent	Percentage increase
employment_days	Duration under the salary

Sample Analytics

- Average salary by department
- Salary growth by year
- Salary distribution by title
- Payroll cost trend

2. FactDepartmentAssignment

Derived from **dept_emp**.

Business Purpose

Analyze employee movement among departments.

Grain

One employee assignment to a department.

```sql
fact_department_assignment
--------------------------
assignment_fact_key (PK)

employee_key (FK)
department_key (FK)

from_date_key (FK)
to_date_key (FK)

assignment_duration_days
```

Measures

Measure	Description
assignment_duration_days	Days in department
assignment_count	1

Sample Analytics

- Employee turnover per department
- Average tenure per department
- Number of employees assigned over time

3. FactDepartmentManager

Derived from **dept_manager**.

Business Purpose

Track managerial assignments.

Grain

One manager assignment per department.

```sql
fact_department_manager
-----------------------
manager_fact_key (PK)

manager_key (FK)
department_key (FK)

from_date_key (FK)
to_date_key (FK)

management_duration_days
```

Measures

Measure	Description
management_duration_days	Duration as manager
manager_count	1

Sample Analytics

- Average manager tenure
- Department leadership history
- Manager turnover analysis

4. FactEmployeeTitleHistory

Derived from **titles**.

Business Purpose

Analyze promotions and career progression.

Grain

One employee title record.

```sql
fact_employee_title_history
---------------------------
title_fact_key (PK)

employee_key (FK)
title_key (FK)

from_date_key (FK)
to_date_key (FK)

title_duration_days
```

Measures

Measure	Description
title_duration_days	Time under title
title_change_count	1

Sample Analytics

- Promotion frequency
- Average time before promotion
- Workforce composition by title

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5. FactWorkforceSnapshot

A periodic snapshot fact table.

Business Purpose

Provide headcount reporting.

Grain

One department snapshot per month.

SQL

```
fact_workforce_snapshot
-----------------------
snapshot_fact_key (PK)

date_key (FK)
department_key (FK)

headcount
male_count
female_count
average_salary
average_tenure
manager_count
```

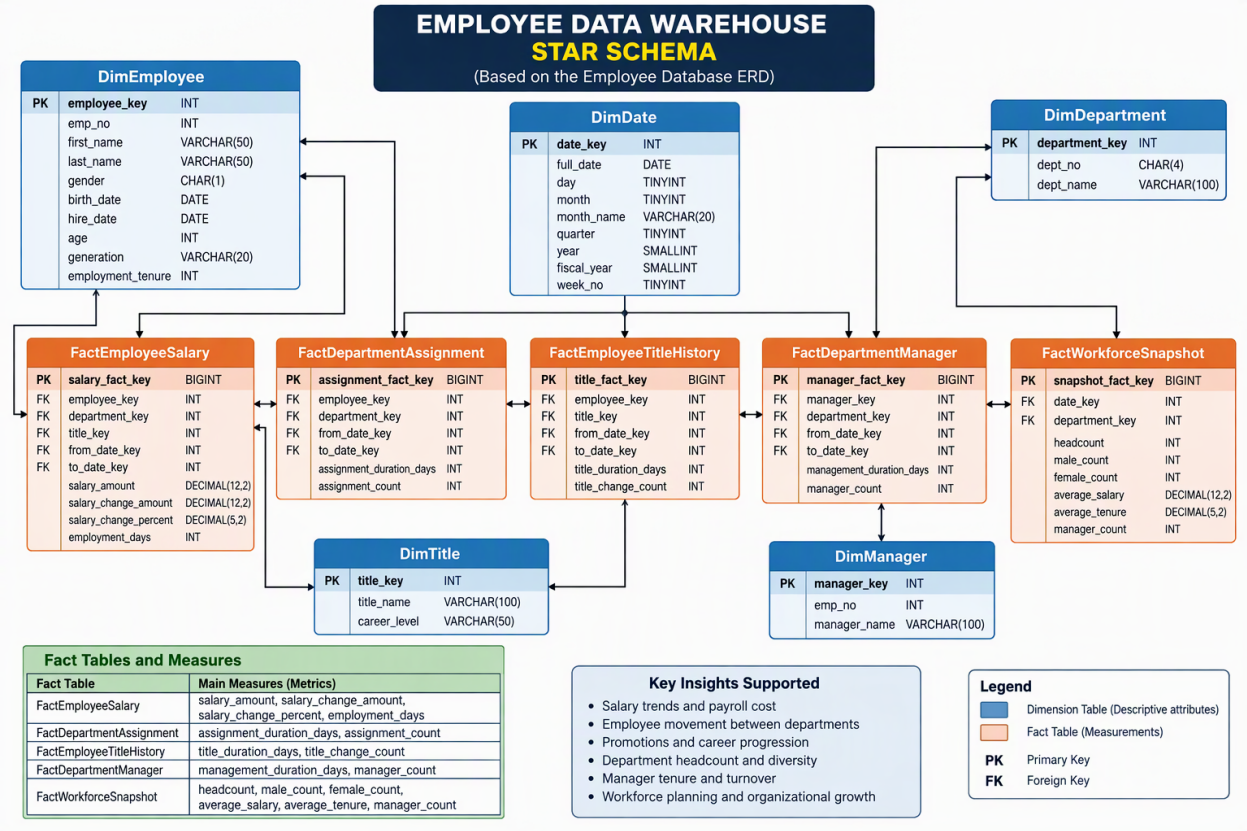
Measures

Measure	Description
headcount	Total employees
male_count	Male employees
female_count	Female employees
average_salary	Average salary
average_tenure	Avg years employed
manager_count	Number of managers

Sample Analytics

- Monthly headcount trends
- Diversity metrics
- Department growth
- Workforce planning

Enterprise Workforce Analytics Star Schema



Activity 2 - Target KPIs

HR KPIs

- Total Employees
- Active Employees

- New Hires
- Employee Turnover Rate
- Average Employee Tenure

Compensation KPIs

- Average Salary
- Median Salary
- Payroll Cost
- Salary Growth Rate
- Department Salary Variance

Career Progression KPIs

- Promotion Rate
- Average Years Before Promotion
- Title Distribution

Organizational KPIs

- Employees per Department
- Span of Control (Employees per Manager)
- Department Growth Rate
- Manager Retention Rate